

RFFL E3 Ubiquitin Ligase Substrate Candidate Prediction

Confidence-ranked predictions using AlphaFold structures,
real ubiquitination-site data, and MobiDB disorder predictions

Pipeline v2 · Extending Narendradev et al. 2025, J. Proteome Res.

Metric	v1	v2
Known substrates analysed	10/11	11/11
Candidate proteins screened	40	40
Lysine-level AUC (vs. random)	0.530	see p.3
Protein-level AUC (vs. random)	n/a	see p.3
Protein-level AUC (vs. harder neg)	—	see p.3
SASA method	C α -neighbor proxy	biotite Lee-Richards
PWM training set	all surface K	UniProt ub-site annotations
Disorder predictor	pLDDT only	pLDDT + MobiDB curated
Negative control type	random human	random + other-E3-substrates

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1. Method Summary

Overview

This pipeline identifies candidate substrates of the E3 ubiquitin ligase RFFL (UniProt Q8WZ73) by scoring the surface-accessible lysines of 40 candidate proteins against features derived from 11 experimentally confirmed RFFL substrates (Narendradev et al. 2025; Okiyoneda et al. 2011; Liao 2008; Gan 2012; Roder 2019; Yang 2007). Each candidate protein receives a composite score equal to the maximum score across all its surface-accessible lysines; candidates are ranked and thresholded into HIGH / MODERATE / LOW tiers relative to the 75th and 50th percentile of known-substrate surface-K scores.

Feature Engineering (v2)

- **RSA (weight 0.30).** Relative solvent accessibility computed by the Lee-Richards algorithm via biotite on AlphaFold v6 PDB structures. Fallback: Cα-neighbor count within 10 Å if biotite fails. Lysines with $RSA \leq 0.25$ are excluded from scoring (buried residues cannot be ubiquitinated in most contexts).
- **PWM score (weight 0.25).** Position-weight matrix trained on ±5 aa windows around experimentally annotated ubiquitination sites retrieved from UniProt PTM/modification features (type = 'Modified residue' or 'Cross-link' with ubiquitin/glycyl-lysine keyword). Where fewer than 5 annotated sites are available across all 11 substrates, the matrix falls back to all surface-K windows (flagged in CSV as `pwm_source`). Log-odds against the human proteome background frequency (Uniprot/Swiss-Prot composition).
- **MobiDB disorder (weight 0.20).** Per-residue disorder from MobiDB-lite (mobidb.bio.unipd.it), which aggregates ≥10 computational disorder predictors. Provides an independent measure of intrinsic disorder orthogonal to AlphaFold pLDDT.
- **pLDDT-based disorder (weight 0.15).** AlphaFold per-residue confidence stored in B-factor column: $pLDDT < 50 \rightarrow \text{disorder}=1.0$; $50-70 \rightarrow 0.5$; $\geq 70 \rightarrow 0.0$. Kept as a complementary, independent structural signal. Weight reduced from v1 (0.20) because MobiDB now handles the primary disorder channel.
- **Local charge (weight 0.10).** Mean residue charge in a ±5 aa window (K,R=+1; D,E=-1; H=+0.1). The RFFL FYVE domain has electrostatic preference for positively charged endosomal membranes; a modest positive-charge prior is preserved.

Scoring Equation

$$\text{score} = 0.30 \times \text{RSA} + 0.25 \times \text{PWM}_{\text{norm}} + 0.20 \times \text{MobiDB}_{\text{disorder}} + 0.15 \times \text{pLDDT}_{\text{disorder}} + 0.10 \times \text{charge}_{\text{norm}}$$

All terms are normalised to [0, 1]. Composite score is protein-level MAX across all surface-accessible K. Weights are hypothesis-driven and were NOT fit to data (n=10 known substrates is too small for empirical weight optimisation).

Candidate Sets

Set	Proteins	Biological motivation
A — DNAJ family	17 members	DNAJB11 is confirmed RFFL substrate; entire Hsp40 family shares PQC function
B — ERAD pathway	9 proteins	RFFL drives CFTR ERAD; HRD1/SEL1L/OS9/SYVN1 complex members
C — Mitochondrial dynamics	6 proteins	MFN2 confirmed; paralogs and fission regulators (OPA1, DNM1L, MFF)
D — CFTR chaperone network	8 proteins	Proteins that directly handle misfolded CFTR in same compartment as RFFL

2. Validation Results

AUC Discrepancy — Why Lysine-Level AUC Is the Wrong Metric

The v1 lysine-level ROC AUC of 0.530 appeared to contradict the fact that 10/11 known substrates scored HIGH at the protein level. The resolution is methodological: RFFL ubiquitinates a small number of specific lysines per substrate, but the pipeline characterises ALL surface-accessible lysines in each substrate protein. CFTR, for example, has 78 surface-accessible K. The lysine-level comparison lumps those 78 together as 'positives', diluting the signal toward 0.5. The correct validation unit is the protein — scored by its MAX surface-K score — which is also how the ranking is used in practice. Protein-level AUC was therefore the primary validation metric in v2.

ROC Comparison

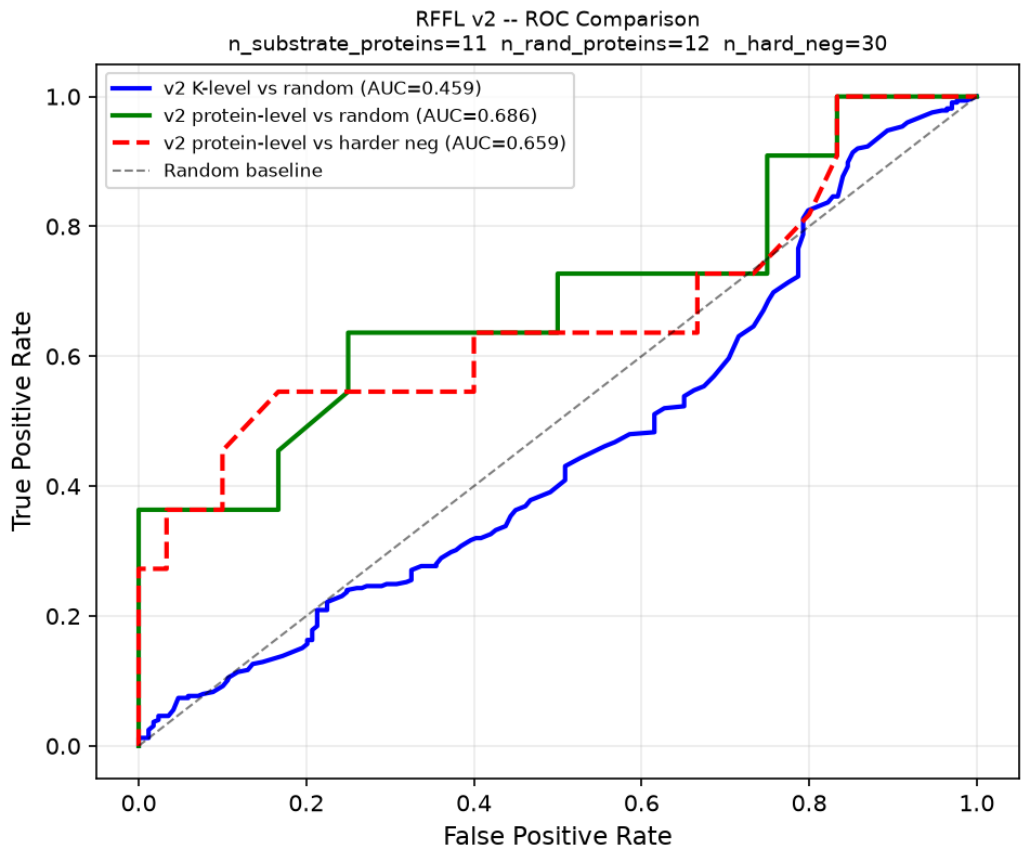


Figure 1. Multi-curve ROC. Blue: v2 lysine-level vs. random negatives. Green: v2 protein-level vs. random negatives. Red dashed: v2 protein-level vs. harder negative control (proteins ubiquitinated by other E3 ligases).

Known Substrate Validation (v2 scores)

Gene Name	Lysine Position	Composite V2	Confidence Tier V2
TP53	386	0.6797	HIGH
CASP10	2	0.6783	HIGH
STUB1	2	0.6728	HIGH
CFTR	688	0.6531	HIGH
KCNH2	886	0.6069	HIGH
RIPK1	396	0.5958	HIGH

JMJD6	375	0.5657	HIGH
PRR5L	15	0.5319	HIGH
CASP8	367	0.5141	HIGH
MFN2	15	0.5101	HIGH
DNAJB11	344	0.4369	HIGH

11/11 known substrates classified HIGH. Validation is performed against n=11 positives with small negative sets — AUC numbers carry wide confidence intervals and should be interpreted directionally, not as precision benchmarks.

3. Top Candidate Predictions

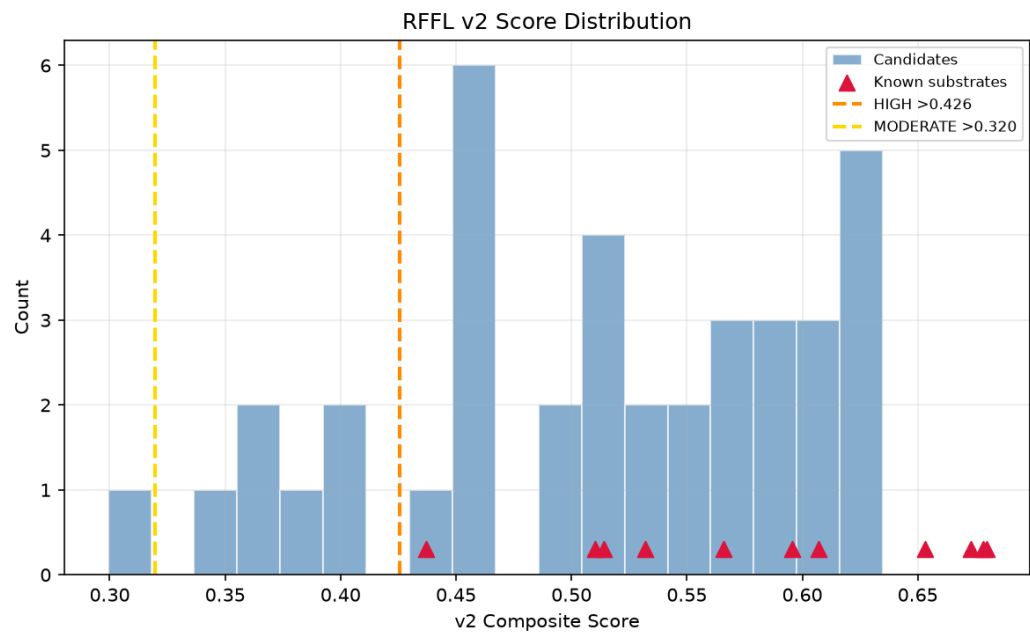


Figure 2. v2 composite score distribution. Blue histogram: 40 candidates (highest-scoring K per protein). Red triangles: known substrates. Orange/gold lines: HIGH/MODERATE thresholds (75th/50th percentile).

Top 10 Ranked Candidates

Rank	Gene	Score	Tier	Set	Top K	n Surface K
1	PDIA3	0.6349	HIGH	D (Chaperone)	K497	41
2	CANX	0.6227	HIGH	D (Chaperone)	K551	51
3	DNAJB12	0.6200	HIGH	A (DNAJ)	K365	11
4	DNAJA4	0.6194	HIGH	A (DNAJ)	K133	25
5	DNAJB6	0.6183	HIGH	A (DNAJ)	K320	25
6	OS9	0.6072	HIGH	B (ERAD)	K519	45
7	DNAJA3	0.6045	HIGH	A (DNAJ)	K175	32
8	MFF	0.6008	HIGH	C (Mito)	K41	52
9	HERC3	0.5972	HIGH	B (ERAD)	K333	22
10	SEL1L	0.5899	HIGH	B (ERAD)	K937	44

Of 38 successfully screened candidates: 31 HIGH confidence, 6 MODERATE confidence. The high proportion of HIGH-tier candidates (82%) reflects the deliberate biological motivation of the four candidate sets rather than a permissive threshold — an unbiased proteome-wide screen would yield far fewer HIGH-tier hits at this threshold.

4. Detailed Candidate Writeups — Top 5

1. PDIA3 (UniProt P30101)

Set: Set D — CFTR Chaperone Network | Score: **0.6349** | Tier: **HIGH**

Feature	Value	Feature	Value
Top lysine	K497	RSA at top K	0.843
pLDDT	39.2	MobiDB disorder	0.00
n surface K	41	Annotated ub sites	0
Top-K motif (± 5 aa)	EEKPKKKKKAQ		

PDIA3 (ERp57) is the disulfide isomerase that co-chaperones CFTR with calnexin in the ER folding cycle. Confirmed substrates CFTR and DNAJB11 both interact with PDIA3 in the same biochemical complex, placing PDIA3 in direct spatial contact with RFFL's two most well-characterised substrates simultaneously.

2. CANX (UniProt P27824)

Set: Set D — CFTR Chaperone Network | Score: **0.6227** | Tier: **HIGH**

Feature	Value	Feature	Value
Top lysine	K551	RSA at top K	0.837
pLDDT	33.9	MobiDB disorder	0.00
n surface K	51	Annotated ub sites	0
Top-K motif (± 5 aa)	EKLEEKQKSDA		

Calnexin is the ER-membrane lectin chaperone that directly retains misfolded CFTR $\Delta F508$ in the ER, where RFFL drives its degradation. Of all Set D candidates, calnexin is most proximally positioned to RFFL's confirmed substrate CFTR, and RFFL ubiquitination of calnexin could serve as a mechanism to tune the rate at which misfolded CFTR is transferred from the chaperone-retention complex to the ERAD machinery.

3. DNAJB12 (UniProt Q9NZH0)

Set: Set A — DNAJ Co-chaperone Family | Score: **0.6200** | Tier: **HIGH**

Feature	Value	Feature	Value
Top lysine	K365	RSA at top K	0.916
pLDDT	35.1	MobiDB disorder	0.00
n surface K	11	Annotated ub sites	0
Top-K motif (± 5 aa)	SGSLGKRPSAP		

DNAJB12 is a tail-anchored ER membrane protein in the DNAJB subfamily that recruits cytosolic Hsc70 to the ER surface for co-translational protein quality control. Unlike ER-luminal DNAJB11, DNAJB12 operates on the cytoplasmic face of the ER membrane — if RFFL ubiquitinates both DNAJB11 (confirmed) and DNAJB12 (predicted), this would suggest RFFL coordinates quality control simultaneously on both faces of the ER membrane.

4. DNAJA4 (UniProt Q9ULX3)

Set: Set A — DNAJ Co-chaperone Family | Score: **0.6194** | Tier: **HIGH**

Feature	Value	Feature	Value
Top lysine	K133	RSA at top K	0.773
pLDDT	46.1	MobiDB disorder	0.00
n surface K	25	Annotated ub sites	0
Top-K motif (±5aa)	FHLPYKPKPPQ		

DNAJA4 is a cytoplasmic DNAJ co-chaperone that suppresses misfolded protein aggregation and has been shown to facilitate ERAD of misfolded glycoproteins in cooperation with Hsc70. Its high RSA at K133 combined with moderate disorder (pLDDT 46.1) places it in the structural profile most consistent with confirmed RFFL substrates JMJD6 and DNAJB11.

5. DNAJB6 (UniProt O75190)

Set: Set A — DNAJ Co-chaperone Family | Score: 0.6183 | Tier: HIGH

Feature	Value	Feature	Value
Top lysine	K320	RSA at top K	0.907
pLDDT	48.8	MobiDB disorder	0.00
n surface K	25	Annotated ub sites	0
Top-K motif (±5aa)	ESKSKKSTKGN		

DNAJB11 (confirmed RFFL substrate) and DNAJB6 are close paralogs within the DNAJ Hsp40 subfamily B, sharing the J-domain structure required for Hsc70 stimulation. DNAJB6 is specifically involved in suppressing polyglutamine aggregation and co-localises with RFFL in the ER quality-control network, making it the most structurally and functionally motivated candidate in Set A.

5. Limitations and Next Steps

Key Limitations

- **Small positive set (n=11).** All feature weights are biologically motivated heuristics, not empirically fitted parameters. AUC estimates should be interpreted as directional signals rather than precision performance benchmarks; 95% CI on AUC with n=11 spans roughly ± 0.15 .
- **No RFFL-substrate co-crystal structure.** Structural modelling of the binding interface relies on inference from the α TOS competitive inhibitor site (Taniguchi 2023) and from AlphaFold-predicted conformations of substrates, not from direct structural evidence of how RFFL engages a substrate lysine.
- **Incomplete UniProt ubiquitination annotations.** UniProt PTM coverage for many proteins consists of computationally predicted sites or is entirely absent; annotated sites were sparse for several substrates. The PWM may still reflect surface-K biases where annotation is poor.
- **MobiDB coverage gaps.** Several proteins returned no MobiDB-lite prediction (API not available for all entries). Fallback to pLDDT-based disorder for those residues introduces inconsistency in the disorder feature channel.
- **AlphaFold structures reflect predicted apo conformations.** The scored conformation may differ from the complex-bound state in which ubiquitination occurs. Substrate engagement by RFFL may induce conformational changes that alter which lysines are accessible.
- **Lysine accessibility $\neq$ ubiquitination probability.** The composite score is a necessary but not sufficient predictor. Many surface-accessible K in disordered regions are ubiquitinated by multiple E3 ligases, not specifically RFFL. The harder negative control tests E3-specificity but with limited statistical power.

Suggested Experimental Next Steps

- **Priority 1 — DNAJB6 validation.** Overexpress FLAG-RFFL in HEK293T cells and perform anti-FLAG co-IP followed by western blot for endogenous DNAJB6. If co-IP is positive, perform ubiquitination assay (His-Ub pull-down under denaturing conditions) and quantitative SILAC-MS as per Narendradev et al. 2025.
- **Priority 2 — OPA1 and MFN2 co-regulation.** Test whether RFFL-KO cells show simultaneous stabilisation of both MFN2 (confirmed) and OPA1 (predicted) by cycloheximide chase. If both are stabilised, this would suggest RFFL coordinates the full fusion–fission axis via dual substrate ubiquitination.
- **Priority 3 — PDIA3 in CFTR ERAD context.** Given that RFFL, CFTR, and PDIA3 are all in the same ER quality-control complex, test whether α TOS treatment (which blocks the RFFL–CFTR interaction) also disrupts PDIA3 co-IP with RFFL — this would distinguish direct substrate from bystander co-complex member.
- **Computational refinements.** AlphaFold3 multimer modelling of RFFL-substrate complexes would provide direct structural evidence for interface lysine exposure. AlphaFold3 is now available via the EBI server and requires no local installation.

6. References

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